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Unit – 5 : Session – 5 : SLO – 2 | Polynomial and Rational Expression Operations

■ I. Questions & Answers

1. Multiply $(x+2)(x+3)$ — `expand((x + 2)*(x + 3))` → $x^2 + 5x + 6$.
2. Factor $x^2 - 4$ — `factor(x**2 - 4)` → $(x - 2)(x + 2)$ (difference of squares).
3. Add $1/(x+1) + 1/(x-1)$ — `together/simplify`: combine over the common denominator $(x+1)(x-1)$ → $((x - 1) + (x + 1)) / (x^2 - 1) = 2x/(x^2 - 1)$.

■ II. Code

- `from sympy import symbols, expand, factor, simplify`
- `x = symbols('x')`
- `print(expand((x + 2)*(x + 3)))` # $x^2 + 5x + 6$
- `print(factor(x**2 - 4))` # $(x - 2)(x + 2)$
- `print(simplify(1/(x + 1) + 1/(x - 1)))` # $2x/(x^2 - 1)$